

Shuo Guo

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Education

The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen) Shenzhen, China
B.Sc. in Applied Mathematics Sept 2025 – May 2029 (expected)
Advisor: Prof. Dr. Zhi-Quan Tom Luo

Research Interests

- Theoretical optimization and the mathematical foundations of machine learning.
- Dynamics, stability, and convergence of stochastic and adaptive optimization algorithms, especially Adam and Muon, with particular interest in dynamical-systems and stochastic-analysis approaches.
- Geometry of flat minima in deep learning optimization.

Experience

Research Assistant, CUHK-Shenzhen May 2026 – Jul 2026
Worked on the empirical evaluation of optimization methods for neural-network training.

Honors and Awards

- Muse Full-tuition Scholarship, CUHK-Shenzhen

Selected Background

- **Convex Optimization:** Studied selected topics from *Lectures on Convex Optimization* and took *Morden Optimization* course by Prof. Dr. Yurii Nesterov, including convexity, gradient methods, strong convexity, complexity bounds, and accelerated first-order methods.
- **Deep Learning and Optimization:** Studied training dynamics and the theoretical analysis of adaptive optimization methods, with particular emphasis on the Adam family of algorithms; read selected chapters of *Mathematical Introduction to Deep Learning: Methods, Implementations, and Theory*.
- **Stochastic Analysis:** Studied measure-theoretic probability and stochastic calculus in course *introduction to stochastic analysis* course, including Brownian motion, stochastic integration, stochastic differential equations, and Itô's formula.
- **Data Science:** Studied selected topics from *Foundations of Data Science*, including SVD, random graphs, concentration, and VC dimension, through an independent course led by Prof. Dr. Zhi-Quan Tom Luo.

Technical Skills

- **Programming:** Python; numerical and machine-learning experiments.
- **Typesetting:** L^AT_EX.

Selected Coursework

- Calculus I and II (Honors);
- Linear Algebra I and II;
- Mathematical Analysis;
- Ordinary Differential Equations (Honors) (taught by Prof. Dr. Weiming Ni);
- Probability I (Honors) (taught by Prof. Dr. Ajay Jasra);
- Introduction to Stochastic Analysis
- Stochastic Differential Equations (taught by Prof. Dr. Dr. Sc. h. c. Michael Röckner, MAE);
- Morden Optimization (taught by Prof. Dr. Yurii Nesterov);
- Independent Study (held by Prof. Dr. Zhi-Quan Tom Luo. We read *Foundations of data science* and held group meetings.)